Probability Notes: Step by Step

Basic to advanced, in the order of Bluman, Elementary Statistics, Chapter 4. Examples anahd exercises are original, built on your lecture notes.

Your Roadmap

Step	Topic	Read in Bluman
1	Experiment, outcome, sample space, tree diagrams	4-1 (p. 183)
2	Events: simple, compound, sure, null	4-1
3	Relations: mutually exclusive, exhaustive, equally likely, complement	4-1, 4-2
4	Classical probability	4-1 (p. 186)
5	Empirical (relative frequency) and subjective probability	4-1 (p. 191)
6	Addition rules (A or B)	4-2 (p. 199)
7	Counting rules: factorial, permutation, combination	4-4 (p. 224)
8	Probability with counting (committee, cards)	4-5 (p. 237)
9	Next level: multiplication rule and conditional probability	4-3 (p. 211)

How to study: read one step, copy the examples by hand, then do that step's exercises before moving on. Answers are at the end. Do not skip Steps 1 to 3. Every later step depends on them.

Step 1: Experiment, Outcome, Sample Space

Why probability?

Some results can be controlled and predicted. For example, marks may depend on study hours and travel hours. But accidents, weather and similar things are uncontrollable, so they are due to **randomness**. Probability is the mathematics used to **model randomness** so we can generalize from a sample to a population.

Definitions

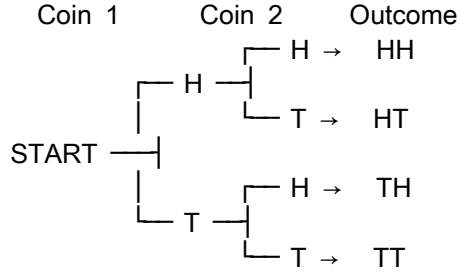
- **Deterministic:** the result is predictable. Example: heating water to 100 °C at sea level makes it boil.
- **Random:** the result is unpredictable, even if the activity is repeated many times under the same conditions. Example: tossing a coin.
- **Probability experiment (random experiment):** an activity that produces results by chance.
- **Trial:** one single performance of the experiment.
- **Outcome:** one possible result of a trial.
- **Sample space (S):** the set of ALL possible outcomes.

Examples

One coin: S = {H, T}

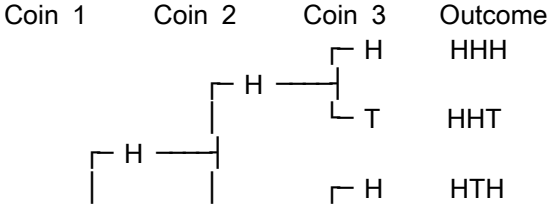
One die: S = {1, 2, 3, 4, 5, 6}

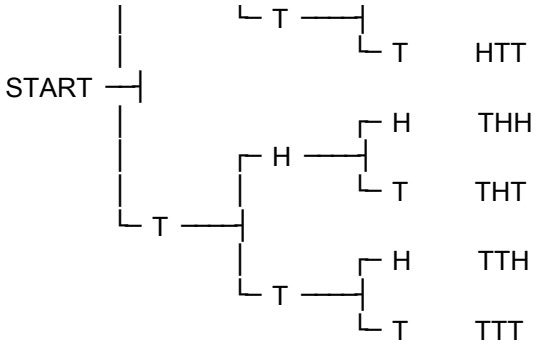
Two coins (tree diagram). Each branch is one coin. Read each path from START to the end to get one outcome.



S = {HH, HT, TH, TT}, so **n(S) = 4**.

Three coins (tree diagram):





n(S) = 8. **Pattern:** each extra coin doubles the outcomes, so k coins give 2^k outcomes.

Two courses (table method). A student gets a grade in MTH262 and in CS (grades A, B, C, D, F). Make a grid, rows = CS grade, columns = MTH262 grade:

CS \ MTH	A	B	C	D	F
A	(A,A)	(A,B)	(A,C)	(A,D)	(A,F)
B	(B,A)	(B,B)	(B,C)	(B,D)	(B,F)
C	...	...	...	...	...
D	...	...	...	...	...
F	...	...	...	...	...

n(S) = 5 × 5 = 25 (the fundamental counting rule, Step 7).

Two dice: 36 outcomes. The number of ways to get each sum:

Sum	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

Exercises: Step 1

1. **(Basic)** Write the sample space for rolling one die and tossing one coin together. How many outcomes?
2. **(Basic)** Deterministic or random? (a) The sum 2 + 3. (b) Which team wins tomorrow's match. (c) The next number on a die. (d) Ice melting at 30 °C.
3. **(Medium)** A family has two children. Draw a tree diagram (B = boy, G = girl) and write S.
4. **(Medium)** How many outcomes are there if a student's grade in two subjects is recorded, using the grades A, B, C, D, F for each?

5. **(Challenge)** How many outcomes are there when 5 coins are tossed? Which pattern did you use?

Step 2: Events

Definitions

- **Event:** any subset of the sample space (a collection of outcomes we care about).
- **Simple event:** an event with exactly ONE outcome. Example: {HT}.
- **Compound event:** an event with MORE than one outcome. Example: {HH, HT}.
- **Sure (certain) event:** the whole sample space S. It always happens.
- **Null (impossible) event:** the empty set ∅. It never happens.

Example: all events for two coins

S = {HH, HT, TH, TT}

- Null event: ∅ (1 event)
- Simple events: {HH}, {HT}, {TH}, {TT} (4 events)
- Compound events with 2 outcomes: {HH,HT}, {HH,TH}, {HH,TT}, {HT,TH}, {HT,TT}, {TH,TT} (6 events)
- Compound events with 3 outcomes: {HH,HT,TH}, {HH,HT,TT}, {HH,TH,TT}, {HT,TH,TT} (4 events)
- Sure event: S (1 event)

Total = 1 + 4 + 6 + 4 + 1 = **16 = 2^4**. If S has n outcomes, there are **2^n possible events**.

Describing events in words

Two coins: 'at least one head' = {HH, HT, TH}. 'No head' = {TT}. 'Exactly one head' = {HT, TH}.

Exercises: Step 2

1. **(Basic)** If S = {a, b, c}, list ALL events. How many are there?
2. **(Basic)** One die: write A = 'even number', B = 'prime number', C = 'number greater

than 6'. Which is the null event?

3. **(Medium)** Three coins: write the event 'exactly two heads' and the event 'at least one tail'. Which one has more outcomes?

4. **(Medium)** Classify as simple, compound, sure or null: (a) rolling a 7 on one die, (b) rolling a number from 1 to 6, (c) rolling a 3, (d) rolling an odd number.

Step 3: Relations Between Events

Definitions

- **Mutually exclusive events:** they cannot happen together, so $A \cap B = \varnothing$. Example: one subject's grade is A or B, never both.
- **Exhaustive events:** together they cover everything, so $A \cup B = S$.
- **Equally likely outcomes:** each outcome has the same chance. A fair coin has H and T equally likely.
- **Complement of A (written A' or not A):** all outcomes of S that are NOT in A.

Warning: exhaustive means $A \cup B = S$ (the whole sample space), not the empty set.

Example: one die

Events	Mutually exclusive?	Exhaustive?	Why
A = {1,2,3}, B = {4,5,6}	Yes	Yes	no common outcome; together they make S
A = even {2,4,6}, B = prime {2,3,5}	No	No	both contain 2; neither contains 1
A = even, B = odd	Yes	Yes	complements of each other

Complements are always mutually exclusive AND exhaustive.

Exercises: Step 3

1. **(Basic)** One die: A = {1,2}, B = {3,4}. Mutually exclusive? Exhaustive?

2. **(Basic)** Name the complement of 'at least one head' in two coin tosses.

3. **(Medium)** In MTH262 you earn one letter grade. Are 'getting A' and 'getting F'

mutually exclusive? Now think of 'getting A in MTH262' and 'getting A in CS': are those mutually exclusive?

4. **(Medium)** Is a weighted die that rolls 6 half the time an equally likely experiment? Explain.

Step 4: Classical Probability

Idea

Probability is a number from 0 to 1 that measures chance. Probability is **objective** when a well-defined method measures the chance, and **subjective** when a person decides it personally. The classical method is objective and works when outcomes are **equally likely**.

Formula

P(A) = n(A) / n(S) = (number of favourable outcomes) / (number of possible outcomes)

Rules (memorize)

1. $0 \le P(A) \le 1$ for every event.
2. $P(S) = 1$ (sure event) and $P(\varnothing) = 0$ (null event).
3. The probabilities of all simple events add up to 1.
4. **Complement rule:** $P(\text{not } A) = 1 - P(A)$.

A probability can NEVER be negative or bigger than 1.

Example 1: two fair coins, one head and one tail

S = {HH, HT, TH, TT}, n(S) = 4. Let A = 'one head and one tail' = {HT, TH}, n(A) = 2.

P(A) = 2/4 = 0.5 = 50%.

Example 2: three coins, exactly H, T, H in that order

n(S) = 8, n(A) = 1, so P(A) = 1/8 = **0.125 = 12.5%**.

Example 3: a card from an ordinary deck (52 cards: 4 suits × 13 ranks; 12 face cards; 26 red, 26 black)

Always define the event first, then use the formula.

Event	n(event)	P
Jack	4	4/52 = 1/13 ≈ 0.077
6 of clubs	1	1/52 ≈ 0.019
Heart	13	13/52 = 1/4
Face card	12	12/52 = 3/13
A 3 or a diamond	4 + 13 − 1 = 16	16/52 = 4/13 (the 3 of diamonds is counted once)

Example 4: complement

Three coins: P(no heads) = 1/8, so P(at least one head) = 1 − 1/8 = 7/8.

Exercises: Step 4

1. **(Basic)** Two fair coins: find P(at least one head).
2. **(Basic)** One die: find P(number greater than 4) and P(prime number).
3. **(Medium)** One card from a deck: P(red card), P(black queen), P(face card or ace).
4. **(Medium)** Two dice: P(sum = 8), P(doubles), P(sum = 1), P(sum between 2 and 12).
5. **(Challenge)** Three coins: find P(exactly two heads) and, using the complement rule, P(at least one tail).

Step 5: Empirical (Relative Frequency) and Subjective Probability

Empirical (relative frequency) approach

Use this when outcomes are NOT equally likely and we must rely on observed data.

P(E) = f / n = (frequency of the class) / (total frequency)

Example: grades of 35 students

Grade	f	Relative frequency
A	8	8/35 ≈ 0.229
B	9	9/35 ≈ 0.257

Grade	f	Relative frequency
C	10	10/35 ≈ 0.286
D	5	5/35 ≈ 0.143
F	3	3/35 ≈ 0.086
Total	35	1 (rounding gives 1.001)

P(getting A) = 0.229. The relative frequencies add up to 1, as they must.

Law of Large Numbers

If an experiment is repeated more and more times, the relative frequency gets closer to the theoretical probability. A fair coin tossed 1000 times might give 520 heads (0.52), and 100,000 tosses will be much closer to 0.50.

Subjective probability

A personal estimate based on experience, opinion or intuition. Example: 'I think there is a 70% chance my team wins.' Nobody can verify it by counting, but it is still used in business and risk decisions.

Summary of the three types

Type	Based on	Example
Classical	equally likely outcomes	P(head) = 1/2
Empirical	observed data	P(A grade) = 8/35
Subjective	personal judgment	'I'm 70% sure it will rain'

(Your lecture also names the **axiomatic definition**: any assignment of probabilities obeying the rules in Step 4 is valid. It is the formal foundation, and Bluman only touches it lightly.)

Exercises: Step 5

1. **(Basic)** Which type is each? (a) P(tail) = 1/2. (b) 12 of 40 customers paid by card, so P = 0.3. (c) 'I believe I have a 90% chance to pass.'
2. **(Medium)** 50 students were asked how they travel to class: bus 20, car 15, bike 10, walk 5. Find P(bike), P(not car), P(bus or walk).
3. **(Medium)** Using the 35-student grade table: find P(B), P(C or better, meaning A, B

or C), P(D or F).

4. **(Challenge)** A coin is tossed 10 times and gives 7 heads. Does this show the coin is unfair? Explain using the Law of Large Numbers.

Step 6: Addition Rules (A or B)

'A or B' means A happens, or B happens, or both.

Rule 1: mutually exclusive events

P(A or B) = P(A) + P(B)

Rule 2: events that are NOT mutually exclusive

P(A or B) = P(A) + P(B) − P(A and B)

We subtract P(A and B) because the overlap was counted twice.

Example 1 (Rule 1): grade table

A and F cannot both happen for one subject, so P(A and F) = 0.

P(A or F) = 8/35 + 3/35 = 11/35 ≈ **0.314**.

Example 2 (Rule 2): heart or ace

P(heart) = 13/52, P(ace) = 4/52, P(ace of hearts) = 1/52.

P(heart or ace) = 13/52 + 4/52 − 1/52 = 16/52 = 4/13.

Example 3: one die, even or prime

Even = {2,4,6}, prime = {2,3,5}, both = {2}.

P = 3/6 + 3/6 − 1/6 = 5/6. Check by listing the union {2,3,4,5,6}: 5 outcomes out of 6. ✓

Exercises: Step 6

1. **(Basic)** One card: P(king or queen).
2. **(Basic)** Two dice: P(sum 7 or sum 11).
3. **(Medium)** One card: P(heart or king).

4. **(Medium)** One die: P(even or greater than 3). Check with the formula AND by listing.
5. **(Challenge)** In a class of 40, 18 take Math, 15 take Stats, and 6 take both. Find P(Math or Stats) and P(neither).

Step 7: Counting Rules

When the sample space is too big to list, we count it.

Fundamental counting rule

If one choice can be made in k_1 ways, a second in k_2 ways, and so on, the total is **$k_1 \times k_2 \times k_3 \times \ldots$**

Example: 3 shirts and 4 pants give 3 × 4 = 12 outfits.

Factorial

n! = n × (n−1) × ... × 2 × 1. For example 5! = 120. By definition **0! = 1**.

Order matters or not?

Take the group {A, B, C} and pick 2.

- **Order does NOT matter (combination):** AB, AC, BC. That is 3 selections, because AB is the same as BA.
- **Order DOES matter (permutation):** AB, BA, AC, CA, BC, CB. That is 6 arrangements. Use this when positions have meaning, such as 1st prize and 2nd prize.

Formulas (choose r from n)

- **Permutation:** nPr = n! / (n − r)!
- **Combination:** nCr = n! / ((n − r)! · r!)

Check: 3P2 = 3!/1! = 6 and 3C2 = 3!/(1!·2!) = 3. ✓

Useful facts: nC0 = 1, nCn = 1, nC1 = n, nCr = nC(n−r). Example: 52C0 = 1.

Quick test: Ask 'does swapping two chosen items give a different result?' If yes, use a permutation. If no, use a combination.

Examples

- 8C3 = 8!/(5!·3!) = (8·7·6)/(3·2·1) = **56**
- 7P2 = 7 · 6 = 42, and 7C2 = 42/2 = 21
- Arrange the 4 letters of MATH: 4! = 24

Exercises: Step 7

1. **(Basic)** Calculate: 5!, 6!/4!, 7P2, 7C2, 10C0, 10C10.
2. **(Basic)** A menu has 3 starters, 4 mains and 2 desserts. How many complete meals?
3. **(Medium)** From 10 students: (a) choose a 3-person committee; (b) choose a president, vice-president and secretary. Which is a combination and which a permutation? Calculate both.
4. **(Medium)** A 4-digit PIN uses digits 0 to 9. How many PINs if repetition is allowed? If no digit may repeat?
5. **(Challenge)** How many different 5-card hands can be dealt from 52 cards?

Step 8: Probability with Counting Rules

The idea is the classical formula, **P(A) = n(A) / n(S)**, but with n(A) and n(S) found using combinations.

Worked example: a committee

A committee of 3 is chosen at random from 5 men and 3 women.

Given: total = 8, men = 5, women = 3, select = 3.

n(S) = 8C3 = **56**.

(a) Exactly 1 man and 2 women. Choose 1 of 5 men and 2 of 3 women:

n(A) = 5C1 × 3C2 = 5 × 3 = 15, so P(A) = 15/56 ≈ **0.268**.

(b) All 3 women. n(B) = 5C0 × 3C3 = 1 × 1 = 1, so P(B) = 1/56 ≈ **0.018**.

(c) At least one woman. The complement is 'all men': 5C3 = 10.

P = 1 − 10/56 = 46/56 ≈ **0.821**.

Method every time:

1. Write the given data and define the event.

2. Find n(S).
3. Find n(event), multiplying the choices from each group.
4. Divide. Check that the answer is between 0 and 1.

Example: two cards

Two cards are drawn. P(both aces) = 4C2 / 52C2 = 6/1326 ≈ 0.0045.

Exercises: Step 8

1. **(Medium)** A committee of 3 is chosen from 6 men and 4 women. Find P(all women), P(exactly 2 men and 1 woman), and P(at least one woman).
2. **(Medium)** Two cards are drawn from a deck. Find P(both hearts).
3. **(Medium)** A bag has 5 red and 3 blue balls and 2 are picked. Find P(both red), P(one of each colour), P(both blue). Do the three answers add up to 1?
4. **(Challenge)** A 5-card hand is dealt. Find P(exactly 2 aces). Hint: 4C2 × 48C3 / 52C5.

Step 9: Next Level (Preview of Bluman 4-3)

Once Steps 1 to 8 are solid, move to events that happen one after another.

- **Independent events:** one does not change the other's probability (draw WITH replacement). **P(A and B) = P(A) × P(B)**
- **Dependent events:** the first changes the second (draw WITHOUT replacement). **P(A and B) = P(A) × P(B | A)**
- **Conditional probability:** the probability of B given A has happened. **P(B | A) = P(A and B) / P(A)**

Example: two cards, both aces.

- With replacement: (4/52)(4/52) = 1/169 ≈ 0.0059.
- Without replacement: (4/52)(3/51) = 1/221 ≈ 0.0045. This matches 4C2/52C2 from Step 8, so the two methods agree.

Exercises: Step 9

1. **(Basic)** Two cards drawn without replacement: P(both kings).

2. **(Basic)** Two coin tosses: P(head then head) using the multiplication rule.
3. **(Medium)** One die: P(6 | the number is even).

Answers

Step 1

1. S = {H1,H2,H3,H4,H5,H6,T1,T2,T3,T4,T5,T6}; 12 outcomes.
2. (a) deterministic, (b) random, (c) random, (d) deterministic.
3. S = {BB, BG, GB, GG}.
4. 5 × 5 = 25.
5. 2^5 = 32 (each coin doubles the outcomes).

Step 2

1. ∅, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}, {a,b,c}; 2^3 = 8 events.
2. A = {2,4,6}, B = {2,3,5}, C = ∅, so C is the null event.
3. 'Exactly two heads' = {HHT, HTH, THH}. 'At least one tail' = all except HHH, which is 7 outcomes, so it has more.
4. (a) null, (b) sure, (c) simple, (d) compound.

Step 3

1. Mutually exclusive: yes. Exhaustive: no (5 and 6 are missing).
2. 'No heads' = {TT}.
3. A and F: yes, mutually exclusive. A in MTH262 and A in CS: no, a student can get both.
4. No. The outcomes do not have equal chances, so the classical formula would not work.

Step 4

1. 3/4 = 0.75.
2. P(>4) = 2/6 = 1/3. P(prime) = 3/6 = 1/2.
3. 26/52 = 1/2. 2/52 = 1/26. Face cards 12 + aces 4 = 16/52 = 4/13.
4. P(sum 8) = 5/36. P(doubles) = 6/36 = 1/6. P(sum 1) = 0. P(sum 2 to 12) = 1.

5. P(exactly two heads) = 3/8. P(at least one tail) = 1 − 1/8 = 7/8.

Step 5

1. (a) classical, (b) empirical, (c) subjective.
2. P(bike) = 10/50 = 0.2. P(not car) = 35/50 = 0.7. P(bus or walk) = 25/50 = 0.5.
3. P(B) = 9/35 ≈ 0.257. P(C or better) = 27/35 ≈ 0.771. P(D or F) = 8/35 ≈ 0.229.
4. Not enough evidence. 10 tosses is a very small sample, and 7 heads can easily happen with a fair coin. The Law of Large Numbers only promises closeness to 0.5 for MANY tosses.

Step 6

1. 8/52 = 2/13 (kings and queens cannot overlap).
2. 6/36 + 2/36 = 8/36 = 2/9.
3. 13/52 + 4/52 − 1/52 = 16/52 = 4/13.
4. Formula: 3/6 + 3/6 − 2/6 = 4/6 = 2/3. Listing: {2,4,5,6} gives 4/6 ✓.
5. P(Math or Stats) = (18 + 15 − 6)/40 = 27/40 = 0.675. P(neither) = 13/40 = 0.325.

Step 7

1. 120; 30; 42; 21; 1; 1.
2. 3 × 4 × 2 = 24.
3. (a) combination: 10C3 = 120. (b) permutation: 10P3 = 720.
4. With repetition: 10^4 = 10,000. Without repetition: 10P4 = 5,040.
5. 52C5 = 2,598,960.

Step 8

1. n(S) = 10C3 = 120. All women: 4C3 = 4, so 4/120 = 1/30 ≈ 0.033. Exactly 2 men and 1 woman: 6C2 × 4C1 = 15 × 4 = 60, so 60/120 = 0.5. At least one woman: 1 − 6C3/120 = 1 − 20/120 = 100/120 ≈ 0.833.
2. 13C2 / 52C2 = 78/1326 = 1/17 ≈ 0.059.
3. n(S) = 8C2 = 28. Both red: 10/28 ≈ 0.357. One of each: 5 × 3 = 15, so 15/28 ≈ 0.536. Both blue: 3/28 ≈ 0.107. Total = 28/28 = 1 ✓.
4. 4C2 × 48C3 = 6 × 17,296 = 103,776. P = 103,776 / 2,598,960 ≈ 0.040.

Step 9

1. (4/52)(3/51) = 1/221 ≈ 0.0045.
2. (1/2)(1/2) = 1/4.
3. P(6 and even) = 1/6, P(even) = 3/6, so P(6 | even) = (1/6)/(3/6) = 1/3.

Where to Get More Practice

Bluman's exercise sets after sections 4-1, 4-2, 4-4 and 4-5 (starting at pp. 183, 199, 224 and 237) match Steps 1 to 8 here. Do the odd-numbered ones first, since the selected answers are at the back of the book.

Part 2: Quick Definitions, Solved Examples, More Exercises

Use this part like this: (1) read the short definitions, (2) read each solved example and cover the solution to try it yourself, (3) do the extra exercises, (4) check the answers at the very end.

Definitions Cheat Sheet (one line each)

- **Random experiment:** an activity whose result is uncertain. **Trial:** one performance of it.
- **Outcome:** one possible result. **Sample space S:** all outcomes. **n(S):** how many.
- **Event:** a subset of S. **Simple:** 1 outcome. **Compound:** 2 or more. **Sure:** S. **Null:** ∅.
- **A ∩ B (A and B):** outcomes in both. **A ∪ B (A or B):** outcomes in A or B or both.
- **Mutually exclusive:** A ∩ B = ∅. **Exhaustive:** A ∪ B = S. **Complement A':** everything not in A.
- **Classical:** P(A) = n(A)/n(S), outcomes equally likely. **Empirical:** P(E) = f/n from data. **Subjective:** personal estimate.
- **Rules:** 0 ≤ P ≤ 1, P(S) = 1, P(∅) = 0, P(A') = 1 − P(A).
- **Addition:** exclusive: P(A or B) = P(A) + P(B). Otherwise: minus P(A and B).
- **Counting:** n! ; nPr = n!/(n−r)! (order matters) ; nCr = n!/((n−r)! r!) (order does not matter).

Step 1 Practice: Sample Space

Solved Example 1.1: A coin is tossed and a die is rolled. Find n(S) and list S.

- *Method:* two stages with 2 choices and 6 choices, so multiply.
- *Solution:* n(S) = 2 × 6 = 12. S = {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}.

Solved Example 1.2: Two dice are rolled. List the outcomes with sum 5.

- *Method:* write pairs (first die, second die) that add to 5.
- *Solution:* (1,4), (2,3), (3,2), (4,1). That is 4 outcomes out of n(S) = 6 × 6 = 36.

Solved Example 1.3: A family has 3 children. How many outcomes for the sexes (B or G)?

- *Method:* 3 stages, 2 choices each: 2 × 2 × 2 = 8.
- *Solution:* S = {BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG}. The tree diagram has the same shape as the 3-coin tree in Part 1.

Extra exercises (Step 1)

1. A spinner has colours R, G, B and is spun twice. List S and find n(S).
2. A coin is tossed 3 times and a die is rolled once. Find n(S).
3. Two dice: list the outcomes with sum 9.
4. A student picks one of 4 courses and one of 3 time slots. How many choices?

Step 2 Practice: Events

Solved Example 2.1: One die. A = odd, B = greater than 4. Find A, B, A ∩ B, A ∪ B.

- A = {1, 3, 5}, B = {5, 6}.
- A ∩ B = {5} (the only common outcome).
- A ∪ B = {1, 3, 5, 6} (list each outcome once).

Solved Example 2.2: Three coins. Write the event 'at least two heads'.

- *Method:* count outcomes with 2 or 3 heads.
- *Solution:* {HHH, HHT, HTH, THH}, which has 4 outcomes, so it is a compound event.

Solved Example 2.3: How many events can be formed if S has 4 outcomes?

- *Solution:* 2^4 = 16 (including ∅ and S).

Extra exercises (Step 2)

1. One die: C = multiples of 3, D = even. Find C ∩ D and C ∪ D.
2. Two dice: write the event 'sum is 10'.
3. How many events does a sample space with 5 outcomes have?
4. Three coins: write 'at most one tail' and say if it is simple or compound.

Step 3 Practice: Relations Between Events

Solved Example 3.1: One die. A = {1,2,3,4}, B = {3,4,5,6}. Mutually exclusive? Exhaustive?

- A ∩ B = {3,4} ≠ ∅, so **not** mutually exclusive.
- A ∪ B = {1,2,3,4,5,6} = S, so **exhaustive**.

Solved Example 3.2: Two dice. Describe the complement of 'sum greater than 9'.

- *Solution:* the complement is 'sum is 9 or less'. Together they cover S and never overlap.

Solved Example 3.3: Two coin tosses. A = at least one head, B = no head. What is the relation?

- A = {HH, HT, TH}, B = {TT}. No overlap, and together they make S, so B is the complement of A.

Extra exercises (Step 3)

1. One die: A = {1,3,5}, B = {2,4,6}. Mutually exclusive? Exhaustive?
2. One die: A = {1,2}, B = {2,3}. Mutually exclusive? Exhaustive?
3. Three coin tosses: name the complement of 'at most 2 heads'.
4. A card is drawn. Are 'heart' and 'king' mutually exclusive? Why?

Step 4 Practice: Classical Probability

Solved Example 4.1: Two dice. Find P(sum = 7).

- *Given:* n(S) = 36.
- *Event A:* sum 7 = {(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)}, n(A) = 6.
- *Solution:* P(A) = 6/36 = 1/6 ≈ 0.167.

Solved Example 4.2: One card. Find P(red face card).

- Red face cards: J, Q, K of hearts and of diamonds = 6.
- P = 6/52 = 3/26 ≈ 0.115.

Solved Example 4.3: Three coins. Find P(at least 2 heads).

- From Example 2.2, n(A) = 4 and n(S) = 8.
- P = 4/8 = 1/2.

Solved Example 4.4 (complement): Two dice. Find P(sum ≠ 7).

- P(sum ≠ 7) = 1 − P(sum = 7) = 1 − 1/6 = 5/6.

Extra exercises (Step 4)

1. Two dice: P(sum ≥ 10). (Use the ways table in Part 1.)
2. One card: P(spade or heart).
3. One die: P(odd number).
4. Three coins: P(all three the same).
5. Two children: P(one boy and one girl).

Step 5 Practice: Empirical Probability

Solved Example 5.1: 40 customers rated a shop: Excellent 12, Good 16, Fair 8, Poor 4. Find P(Good), P(Fair or Poor), P(at least Good).

- P(Good) = 16/40 = 0.40.
- P(Fair or Poor) = (8 + 4)/40 = 12/40 = 0.30.
- P(at least Good) = (12 + 16)/40 = 28/40 = 0.70.

Solved Example 5.2: 50 households and their number of cars: 0 cars: 5, 1 car: 20, 2 cars: 18, 3 cars: 7. Find P(2 cars), P(at least 1 car), P(more than 1 car).

- P(2) = 18/50 = 0.36.
- P(at least 1) = 1 − 5/50 = 0.90.

- P(more than 1) = (18 + 7)/50 = 0.50.

Solved Example 5.3: Which type of probability? (a) 'The chance my exam is hard is 80%, I feel.' (b) 15 of 500 tested bulbs were faulty. (c) A fair die shows 4.

- (a) subjective, (b) empirical (0.03), (c) classical (1/6).

Extra exercises (Step 5)

1. In a survey of 200 people, 120 prefer tea and 80 prefer coffee. Find P(tea) and P(coffee).
2. 500 items were tested and 15 were defective. Find P(defective) and P(not defective).
3. It rained on 9 of 30 days. Estimate P(rain on a given day).
4. Name the type: 'The manager says the new product has a 60% chance of success.'

Step 6 Practice: Addition Rules

Solved Example 6.1 (mutually exclusive): One card. P(king or ace).

- A king cannot be an ace, so no overlap.
- P = 4/52 + 4/52 = 8/52 = 2/13.

Solved Example 6.2 (not exclusive): One card. P(spade or face card).

- P(spade) = 13/52, P(face) = 12/52, P(spade and face) = 3/52 (J, Q, K of spades).
- P = 13/52 + 12/52 − 3/52 = 22/52 = 11/26 ≈ 0.423.

Solved Example 6.3: Two dice. P(sum 5 or doubles).

- P(sum 5) = 4/36, P(doubles) = 6/36.
- Overlap: a sum of 5 is odd, but a double has an even sum, so P(both) = 0.
- P = 4/36 + 6/36 = 10/36 = 5/18.

Solved Example 6.4: In a town, 60% of people drink tea, 50% drink coffee, and 25% drink both. Find P(tea or coffee).

- P = 0.60 + 0.50 − 0.25 = 0.85.

Extra exercises (Step 6)

1. One card: P(red or queen).

2. One die: P(odd or less than 3). Check by listing.
3. Two dice: P(sum 6 or sum 8).
4. Of 100 students, 60 play cricket, 45 play football and 25 play both. Find P(either game), P(neither), and P(cricket only).

Step 7 Practice: Counting Rules

Solved Example 7.1: Compute 6! and 9!/7!.

- 6! = 720.
- 9!/7! = 9 × 8 = 72 (cancel 7! from top and bottom).

Solved Example 7.2: Compute 8P3 and 8C3.

- 8P3 = 8 × 7 × 6 = 336.
- 8C3 = 336 / 3! = 336/6 = 56.
- Note: nCr = nPr / r!.

Solved Example 7.3 (fundamental rule): A licence plate has 2 letters followed by 3 digits, repetition allowed. How many plates?

- 26 × 26 × 10 × 10 × 10 = 676,000.

Solved Example 7.4: (a) Choose a 4-person team from 12. (b) Arrange 5 books on a shelf.

- (a) Order does not matter: 12C4 = (12·11·10·9)/(4·3·2·1) = 495.
- (b) Order matters: 5! = 120.

Extra exercises (Step 7)

1. Compute 9C2 and 6P4.
2. In a race of 5 runners, how many ways can the top 3 places (gold, silver, bronze) be filled?
3. From 4 men and 3 women, how many ways to choose 2 men and 1 woman?
4. How many ways can 3 prizes of equal value be given to 3 of 8 students?

Step 8 Practice: Probability with Counting

Solved Example 8.1: A box has 10 items, 3 of them defective. 4 items are picked at random. Find P(none defective), P(exactly 1 defective), P(at least 1 defective).

- *Given:* total 10, defective 3, good 7, pick 4. n(S) = 10C4 = 210.
- None defective: 7C4 = 35, so P = 35/210 = 1/6 ≈ 0.167.
- Exactly 1 defective: 3C1 × 7C3 = 3 × 35 = 105, so P = 105/210 = 0.5.
- At least 1: 1 − 1/6 = 5/6 ≈ 0.833.

Solved Example 8.2: A 5-card hand is dealt. Find P(all five are clubs).

- n(S) = 52C5 = 2,598,960. n(A) = 13C5 = 1,287.
- P = 1,287 / 2,598,960 ≈ 0.0005.

Solved Example 8.3: A class has 12 boys and 8 girls. 4 are chosen for a quiz team. Find P(2 boys and 2 girls).

- n(S) = 20C4 = 4,845. n(A) = 12C2 × 8C2 = 66 × 28 = 1,848.
- P = 1,848 / 4,845 ≈ 0.381.

Extra exercises (Step 8)

1. A bag has 4 red and 6 green balls and 3 are picked. Find P(all green), P(2 red and 1 green), P(at least one red).
2. Two cards are drawn. Find P(one ace and one king).
3. A lottery asks you to pick 6 numbers from 20. Find the chance of picking the winning set.
4. A 5-card hand is dealt. Find P(exactly 3 kings).

Answers to Part 2 Extra Exercises

Step 1

1. S = {RR, RG, RB, GR, GG, GB, BR, BG, BB}; n(S) = 9.
2. $2^3 \times 6 = 48$.
3. (3,6), (4,5), (5,4), (6,3).
4. 4 × 3 = 12.

Step 2

1. C = {3,6}, D = {2,4,6}. C ∩ D = {6}; C ∪ D = {2,3,4,6}.
2. {(4,6), (5,5), (6,4)}.
3. 2^5 = 32.
4. {HHH, HHT, HTH, THH}. Compound (4 outcomes).

Step 3

1. Mutually exclusive: yes. Exhaustive: yes (they are complements).
2. Mutually exclusive: no (both contain 2). Exhaustive: no (4, 5, 6 are missing).
3. 'Exactly 3 heads' = {HHH}.
4. No. The king of hearts is both a heart and a king.

Step 4

1. Sums 10, 11, 12 have 3 + 2 + 1 = 6 ways, so 6/36 = 1/6.
2. 26/52 = 1/2.
3. 3/6 = 1/2.
4. {HHH, TTT}, so 2/8 = 1/4.
5. S = {BB, BG, GB, GG}; favourable = {BG, GB}, so 2/4 = 1/2.

Step 5

1. P(tea) = 120/200 = 0.6; P(coffee) = 80/200 = 0.4.
2. P(defective) = 15/500 = 0.03; P(not) = 0.97.
3. 9/30 = 0.30.
4. Subjective.

Step 6

1. 26/52 + 4/52 − 2/52 = 28/52 = 7/13.
2. 3/6 + 2/6 − 1/6 = 4/6 = 2/3. Listing: {1,2,3,5} gives 4/6 ✓.
3. 5/36 + 5/36 = 10/36 = 5/18.
4. P(either) = (60 + 45 − 25)/100 = 0.80; P(neither) = 0.20; P(cricket only) = (60 − 25)/100 = 0.35.

Step 7

1. 9C2 = 36; 6P4 = 6·5·4·3 = 360.

2. 5P3 = 60 (order matters: gold, silver, bronze differ).
3. 4C2 × 3C1 = 6 × 3 = 18.
4. Equal prizes means order does not matter: 8C3 = 56.

Step 8

1. n(S) = 10C3 = 120. All green: 6C3 = 20, so 20/120 ≈ 0.167. 2 red and 1 green: 4C2 × 6C1 = 36, so 36/120 = 0.3. At least one red: 1 − 20/120 ≈ 0.833.
2. n(S) = 52C2 = 1,326. n(A) = 4 × 4 = 16. P = 16/1,326 ≈ 0.012.
3. 20C6 = 38,760, so the chance is 1/38,760 ≈ 0.000026.
4. 4C3 × 48C2 = 4 × 1,128 = 4,512. P = 4,512 / 2,598,960 ≈ 0.0017.

Study Tip

For every probability problem, do the same four things: (1) write what is given, (2) define the event, (3) find n(S) and n(event), (4) divide and check that the answer is between 0 and 1. If a problem says 'at least', try the complement first.